Arunabha Majumder

Mechanical | Mechatronics Engineer

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Siemens Energy, Gas Services, Finspang

Application: Structural Calculation Engineer (id 287947)

Dear Hiring Team,

I am applying for the Structural Calculation Engineer role in Finspang. I am a mechanical engineer doing my PhD in Mechatronics and Robotics at Aalborg University, with my thesis expected to be finished in March. What I enjoy most in my work is the part where a model meets a measurement, and you find out if your numbers were right. That is a big part of this role, and it is why I am applying.

In my PhD I worked on stiffness modelling and Jacobian-based force estimation for a 5-bar parallel robot with variable-stiffness actuators. I checked the estimated forces against a force sensor and got them within 10% RMS error. I also designed the load-carrying parts of a shoulder exoskeleton, checked them with stiffness analysis in SolidWorks and optimised the design with a genetic algorithm. I built test rigs and integrated motors, encoders and force sensors to measure how the system really behaved. I use MATLAB nearly every day for calculations and processing test data, and I have basic Python.

I want to be honest about the main gap. Apart from stiffness analysis in SolidWorks, I have not used dedicated tools like Abaqus or Ansys so far. My base in mechanics, modelling and validation is solid, and building deeper simulation skills is what I most want to do next. I learn new tools fast when I can see how they connect to real hardware and test results, and I think a team of experienced calculation engineers is the best place to learn it properly.

I am also comfortable explaining technical results. I have written conference papers, taught university courses and supervised students, and I have a Diploma in Fine Arts, so I make clear figures and visuals for reports and presentations.

Thank you for considering my application. My portfolio is at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder